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4.7 Magnetism and Electromagnetism

AQA GCSE Physics (8463) — Paper 2

4.7.1 Poles of a Magnet

A **permanent magnet** is always magnetic due to the alignment of its atomic electrons. Examples: steel, iron alloys, natural magnetite. A **magnetic pole** is the region of a magnet where the magnetic field is strongest.

Types of Magnets:

- **Permanent magnets:** Always magnetic (e.g. steel, iron alloys, cobalt). The magnetism cannot be switched off.
- **Induced magnets:** Only magnetic when placed near a permanent magnet (e.g. iron, cobalt, nickel). The magnetism is temporary and disappears when removed from the field.

Every magnet has exactly two poles: a **North pole (N)** and a **South pole (S)**. Magnetic field lines emerge from the North pole and enter at the South pole.

Magnetic Poles and Forces

The Law of Magnetic Poles:

- **Like poles repel:** North–North and South–South poles push apart.
- **Unlike poles attract:** North–South poles pull towards each other.
- These are **non-contact forces** — the magnets do not need to touch.

Induced magnets always experience an **attractive force** regardless of which pole of the permanent magnet is nearest. This is because the induced magnet's poles automatically align to attract the permanent magnet.

Exam Tip: Remember: like poles REPEL, unlike poles ATTRACT. Induced magnets ALWAYS attract (never repel). This distinction is key in exam questions about iron filings or compass needles.

4.7.2 Magnetic Fields

A **magnetic field** is the region of space around a magnet (or current-carrying wire) where another magnet or magnetic material experiences a non-contact force.

Magnetic Field Lines and Direction

Key properties of magnetic field lines:

- Field lines always point from North pole to South pole **outside the magnet** (and South to North **inside** the magnet).
- The direction of a field line shows the direction of the force on a North pole placed at that point.
- **Field lines never cross.**
- **Closer field lines indicate a stronger field;** the field is strongest near the poles.

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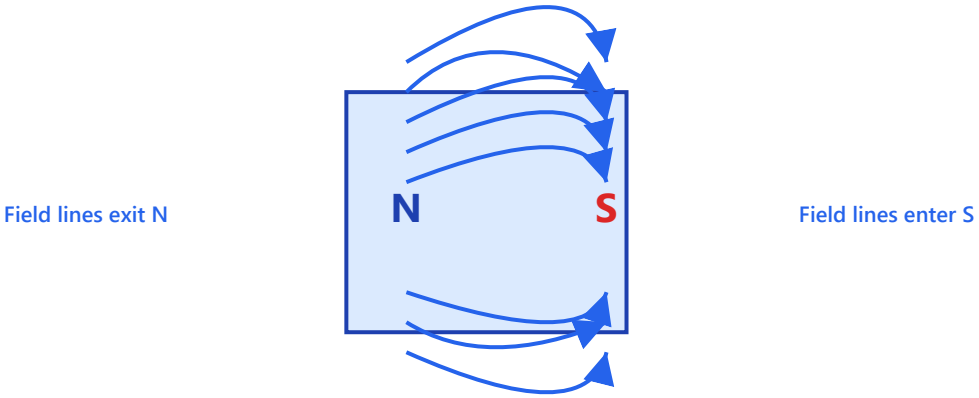
Magnetic Field Around a Bar Magnet



Plotting Magnetic Fields

To map a magnetic field, use a **plotting compass:** a small magnetized needle that aligns with the magnetic field direction. Place the compass at various points, mark the direction the needle points, and trace a continuous line through all points. The resulting curve is a field line.

> **Exam Tip:** Magnetic field lines are always drawn from N pole to S pole outside the magnet. Stronger fields have closer-spaced lines. Never let lines cross — each point has only one field direction.

4.7.3 Earth's Magnetic Field

> **Earth's magnetic field** is generated by the movement of molten iron in Earth's outer core. The field is approximately that of a giant bar magnet, tilted relative to Earth's rotational axis.

Key facts about Earth's magnetic field:

- The magnetic North Pole is not at the geographic North Pole — it is offset by about 10° and drifts slowly over time.
- A **compass needle** is a small permanent magnet that aligns with Earth's magnetic field and points roughly towards magnetic North.

- Compasses work anywhere on the globe, proving that Earth has a global magnetic field.
- The field extends far into space and is constantly regenerated by currents in the molten iron core.

> Evidence for Earth's magnetic field: compasses align with a consistent direction nearly everywhere on Earth's surface. This behaviour proves that Earth possesses a large-scale, persistent magnetic field generated internally.

> **Exam Tip:** Earth's magnetic field is why compasses work. Remember: the field is generated by molten iron movement in the core, not by any external source. The magnetic poles drift slowly over geological timescales.

4.7.4 Magnetic Field Around a Current-Carrying Wire

> An **electric current flowing through a wire creates a magnetic field** around the wire. The stronger the current and the closer to the wire, the stronger the field.

Direction of the Field

The direction of the magnetic field around a straight current-carrying wire can be found using the **right-hand grip rule:**

Right-hand grip rule: Point your right thumb in the direction of conventional current flow along the wire. Your fingers curl in the direction of the magnetic field lines around the wire.

Strength of the Field

The magnetic field around a current-carrying wire:

- **Stronger with larger current** — doubling the current doubles the field strength.
- **Stronger closer to the wire** — the field weakens rapidly with distance (inversely proportional to distance).
- **Reverses if current direction reverses** — reversing current reverses the field direction.

> **Exam Tip:** Use the right-hand grip rule in exams: thumb = current, fingers = field direction. If asked to draw field lines around a wire, draw concentric circles (when viewed end-on) with no crossings. If current reverses, all field directions reverse.

4.7.5 Solenoid (Coil)

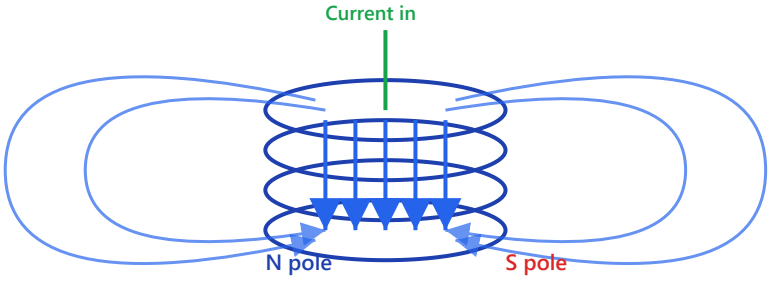
> A **solenoid** is a cylindrical coil of insulated wire wound tightly in a helix. When current passes through it, a strong uniform magnetic field is produced inside, resembling a bar magnet's field.

Field Inside and Outside a Solenoid

Inside: The field is approximately uniform (parallel, equally-spaced lines) and strong, pointing along the axis from one end to the other.

Outside: The field resembles that of a bar magnet — field lines spread out from one pole and curve around to enter at the other pole.

Magnetic Field Inside and Outside a Solenoid



Electromagnets

> An **electromagnet** is a solenoid with an iron (or soft-iron) core inserted through its centre. The iron core greatly amplifies the magnetic field by concentrating and aligning the field lines.

Advantages of electromagnets over permanent magnets:

- **Switchable:** Turn current on/off to control magnetism instantly.
- **Variable strength:** Increase current or add more turns to increase field strength.
- **No permanent demagnetization:** Soft iron does not retain magnetism after current is cut.

Practical Uses of Electromagnets

- **Lifting magnets (cranes):** Lift cars and scrap metal at recycling yards.
- **Maglev trains:** Magnetic levitation uses electromagnets to suspend and propel trains.
- **Loudspeakers:** Coil in electromagnet field vibrates to produce sound.
- **Electric motors:** Electromagnets create torque to rotate mechanical loads.
- **Magnetic door locks:** Engage and release using electromagnets.
- **Doorbells and relays:** Electromagnets switch circuits on and off.

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Exam Tip: Electromagnets require: (1) a solenoid, (2) an iron core, (3) a current. Increasing current or turns increases field strength. They are the basis of motors, speakers, and many industrial applications.

4.7.6 Motor Effect

The **motor effect** (or **Lorentz force**) occurs when a current-carrying conductor is placed in a magnetic field: the conductor experiences a force perpendicular to both the current direction and the magnetic field direction.

Motor Effect Formula

The magnitude of the force on a current-carrying conductor in a magnetic field is given by:

$$F = B \times I \times L$$

where:

- **F** = force (N, newtons)
- **B** = magnetic flux density (T, tesla)
- **I** = current (A, amperes)
- **L** = length of conductor in the field (m, metres)

Units and Definitions

Tesla (T): The unit of magnetic flux density. 1 tesla = 1 newton per ampere per metre (1 N/(A·m)).

Worked Examples

Example 1: A wire of length 0.5 m carries a current of 3 A in a magnetic field of flux density 0.2 T. Calculate the force on the wire.

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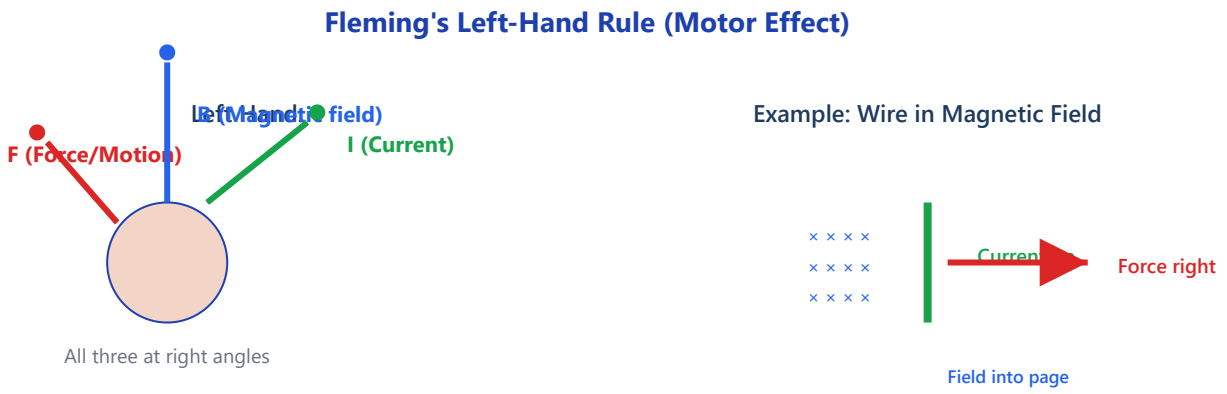
```
F = B × I × L
F = 0.2 × 3 × 0.5
F = 0.3 N
```

Example 2: A conductor experiences a force of 10 N in a field of 2 T with a current of 5 A. Find the length of conductor in the field.

```
F = B × I × L
10 = 2 × 5 × L
L = 10 / (2 × 5) = 1 m
```

Fleming's Left-Hand Rule

To find the direction of the force (or motion) on a current-carrying conductor in a magnetic field, use **Fleming's left-hand rule:**



How to use Fleming's left-hand rule:

1. Point your **first finger** in the direction of the **Magnetic field (B)** (usually from N to S pole).
2. Point your **second finger** in the direction of the electric **Current (I)**.
3. Your **thumb** then points in the direction of the **Force (F)** or motion of the conductor.

4. All three must be at right angles to each other.

Exam Tip: Memorize: Left hand, Lorentz force. Thumb = Force, First finger = Field, seCond finger = Current. Use this to predict wire motion in exam diagrams. If current or field reverses, the force direction reverses.

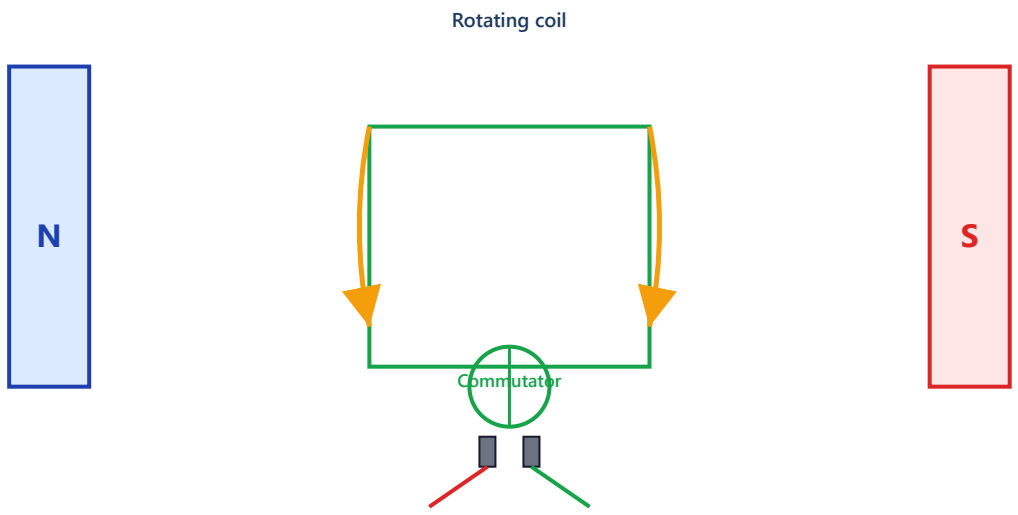
4.7.7 Electric Motor

A **d.c. electric motor** converts electrical energy into mechanical motion by using the motor effect on a rotating coil in a magnetic field.

How a D.C. Motor Works

Basic principle: A rectangular coil of wire carries current in a magnetic field between two permanent magnets (or electromagnets). The motor effect produces forces on opposite sides of the coil in opposite directions, creating a turning moment (torque) that rotates the coil.

Simple D.C. Motor



The Split-Ring Commutator

The **split-ring commutator** is a split metal ring attached to the coil that reverses the direction of current through the coil every half turn. This ensures the force on each side of the coil always points in the same rotational direction, keeping the coil spinning continuously.

How it works:

- As the coil rotates, the commutator switches which half of the ring is in contact with each brush (carbon contact).
- When the current switches, the force direction on opposite sides of the coil reverses, but both forces now push in the direction of ongoing rotation.
- Without the commutator, the coil would oscillate back and forth instead of rotating continuously.

Carbon brushes are sliding electrical contacts that maintain connection between the stationary power supply and the rotating commutator. They conduct current reliably despite the commutator's rotation.

Practical Applications

- **Electric fans:** Motor drives blades to circulate air.
- **Electric cars:** Motors provide propulsion force.
- **Washing machines:** Motor rotates the drum.
- **Blenders:** Motor spins blades to mix ingredients.
- **Drills:** Motor drives the drill bit.

Exam Tip: Remember: the split-ring commutator reverses current direction every half turn to maintain continuous rotation in the same direction. Without it, the coil would oscillate. Carbon brushes maintain electrical contact with the rotating commutator.

4.7.8 Loudspeakers and Microphones (Triple Only)

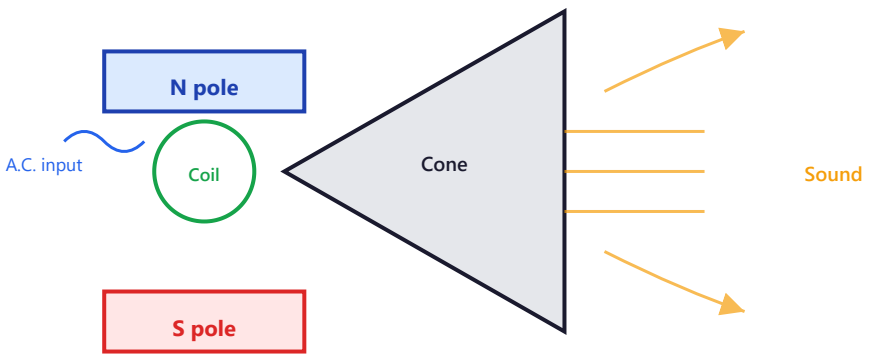
Loudspeakers

A **loudspeaker** converts electrical audio signals (a.c. current) into sound waves using the motor effect.

How a loudspeaker works:

- A coil is attached to a flexible cone and positioned in a strong radial magnetic field (from a permanent magnet).
- As alternating current flows through the coil, the motor effect produces a varying force on the coil in the direction of the magnetic field axis.
- The coil oscillates back and forth, pushing and pulling the cone.
- The moving cone compresses and rarefies the air, creating pressure waves → sound waves that travel to the listener's ear.

Loudspeaker Cross-Section



Microphones

A **microphone** converts sound waves (pressure waves in air) into electrical audio signals. It uses the generator effect (electromagnetic induction) to produce a varying potential difference that mirrors the sound wave.

How a microphone works:

- Sound waves from a source strike a flexible diaphragm, causing it to vibrate back and forth.

- A coil attached to the diaphragm moves in a fixed magnetic field.
- As the coil moves through the field (generator effect), a varying potential difference is induced in the coil.
- The frequency and amplitude of the induced voltage match the frequency and amplitude of the sound, creating an electrical audio signal.
- This signal can be amplified and transmitted (e.g., to a speaker or recording device).

Exam Tip: Remember: loudspeaker (electrical → mechanical → sound), microphone (sound → mechanical → electrical). Both use coils in magnetic fields but in opposite directions of energy conversion.

4.7.9 Induced Potential, Generators, and Lenz's Law (Triple Only)

The **generator effect** is the production of a potential difference (and hence a current) when a conductor moves relative to a magnetic field, or when a magnetic field changes through a conductor/coil.

Conditions for Inducing Voltage

A potential difference is induced when there is a **change in magnetic flux** through a conductor or coil. This can happen by:

- **Moving a conductor in a magnetic field:** sliding a rod perpendicular to field lines.
- **Moving a coil in a magnetic field:** rotating a coil between magnet poles.
- **Changing the field strength:** increasing or decreasing the field through a fixed coil.
- **Changing the area of the coil:** deforming or moving a coil to alter its exposure to the field.

Lenz's Law (Concept Only)

Lenz's law: The direction of an induced potential difference (and hence induced current) is such that it opposes the change that caused it. The induced effect always works against the

> cause.

Practical meaning: When a magnet approaches a coil, the induced current creates a magnetic field that repels the magnet. This opposes the magnet's motion, requiring work to be done to move the magnet. This is why magnetic braking and eddy current damping are possible.

Ways to Increase Induced Voltage

The induced voltage depends on:

- **Faster motion or faster field change:** Moving a conductor faster through a field increases the rate of flux change → higher induced voltage.
- **Stronger magnetic field:** A stronger field (higher B) increases the magnitude of flux change → higher voltage.
- **More turns on the coil:** A coil with more loops has more flux linkage → higher voltage (each turn adds to the total induced voltage).
- **Larger coil area:** A bigger coil experiences more flux change → higher voltage.

4.7.10 A.C. Generator (Alternator) and D.C. Dynamo (Triple Only)

A.C. Generator (Alternator)

> An **A.C. generator (alternator)** is a device that produces alternating current by rotating a coil in a uniform magnetic field. It uses slip rings to maintain continuous electrical contact while the coil rotates.

How an A.C. generator works:

- A rectangular coil is rotated in a magnetic field between two permanent magnets (or electromagnets).
- As the coil rotates, the magnetic flux through it changes continuously (increasing, then decreasing, then reversing).
- This changing flux induces a sine-wave voltage across the coil (generator effect).

- The voltage alternates: positive half-cycle, then negative half-cycle, with a frequency equal to the rotation frequency.
- **Slip rings:** Two complete rings that rotate with the coil. Carbon brushes maintain electrical contact, allowing the alternating voltage to be drawn from the rotating coil.

D.C. Dynamo

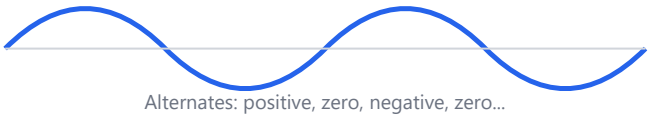
> A **d.c. dynamo** is a generator that produces direct current by rotating a coil in a magnetic field, using a split-ring commutator to rectify the current direction.

How a d.c. dynamo works:

- Similar to an A.C. generator, but uses a **split-ring commutator** instead of slip rings.
- The commutator reverses the external connections every half rotation, so the current always flows in the same direction externally (even though the coil's position reverses).
- The output is d.c., but with varying magnitude (a bumpy d.c. waveform) rather than a smooth sine wave.
- This is rectified d.c. — full-wave rectified output.

A.C. vs D.C. Output Waveforms

A.C. Generator Output (Sine Wave)



Alternates: positive, zero, negative, zero...

D.C. Dynamo Output (Rectified)

Always positive, varying magnitude

4.7.11 Transformers (Triple Only)

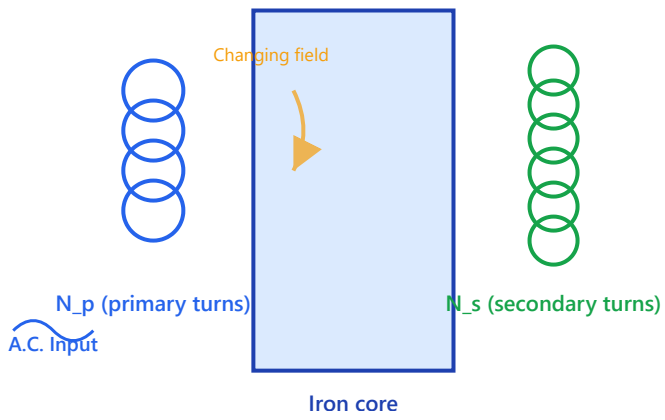
A **transformer** is a device that changes the voltage of an alternating current while keeping the power approximately constant. It consists of two coils (primary and secondary) wound on a common iron core.

How Transformers Work

Basic operation:

- Alternating current in the primary coil creates a constantly-changing magnetic field in the iron core.
- This changing field passes through the secondary coil, inducing an alternating voltage (transformer equation).
- The voltage ratio depends only on the ratio of turns: more turns on secondary → higher voltage (step-up); fewer turns → lower voltage (step-down).

Simple Transformer



Transformer Equations

Voltage ratio: `V_p / V_s = N_p / N_s`

where:

- **V_p** = primary voltage (input)
- **V_s** = secondary voltage (output)
- **N_p** = number of turns on primary coil
- **N_s** = number of turns on secondary coil

Step-up transformer: N_s > N_p → V_s > V_p (voltage increased, current decreased)

Step-down transformer: N_s < N_p → V_s < V_p (voltage decreased, current increased)

Power conservation (assuming 100% efficiency): `V_p × I_p = V_s × I_s`

Power input = Power output. If voltage increases, current must decrease proportionally.

Worked Examples

Example 1: A step-up transformer has 100 turns on the primary and 1000 turns on the secondary. The primary voltage is 10 V. Calculate the secondary voltage.

```
V_p / V_s = N_p / N_s
10 / V_s = 100 / 1000
V_s = 10 × (1000 / 100) = 100 V
```

Example 2: A transformer has a primary voltage of 230 V and a secondary voltage of 12 V. The primary current is 0.5 A. Calculate the secondary current (assuming 100% efficiency).

```
V_p × I_p = V_s × I_s
230 × 0.5 = 12 × I_s
I_s = 115 / 12 ≈ 9.6 A
```

Key Limitation

Transformers only work with alternating current (A.C.). A constantly-changing (alternating) voltage is needed to induce a changing magnetic field. With d.c., the field is constant, so no

voltage is induced in the secondary coil.

Exam Tip: Transformers are 100% efficient only theoretically. Real transformers have losses due to resistance in coils and eddy currents in the core. Always use the power equation $V_p \times I_p = V_s \times I_s$ unless told otherwise in the exam.

4.7.12 The National Grid (Triple Only)

The **National Grid** is the system of power stations, transformers, and cables that distributes electrical power from generating stations to homes and businesses across the country.

Why High Voltage Transmission?

The problem: Electrical power is transmitted over long distances (hundreds of km) from power stations to towns and cities. If transmitted at low voltage, the current would be huge, and power losses in the cables would be enormous.

The solution: Use transformers to step up the voltage to very high levels (e.g. 400 kV) for transmission. This reduces the current, dramatically cutting power losses.

Power Loss in Transmission

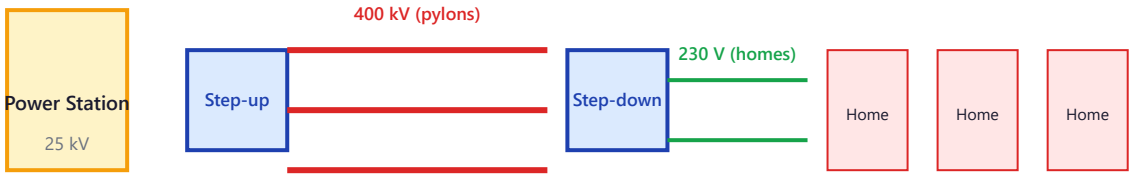
Power lost in cables: $P_{loss} = I^2 \times R$

where R is the resistance of the transmission cable. Halving the current reduces power loss to 1/4 (since I^2 is halved, so I^2 is quartered).

The National Grid System

Simplified pathway:

Simplified National Grid



1. **Power station (25 kV):** Generates electricity at ~25,000 V.
2. **Step-up transformer:** Increases voltage to ~400,000 V (400 kV) for transmission over long distances.
3. **Transmission lines (pylons):** High-voltage overhead cables carry power across the country with minimal loss.
4. **Step-down transformer (substation):** Reduces voltage to safer, lower levels (~230 V) for use in homes.
5. **Local distribution and homes:** Domestic appliances use 230 V a.c. supply.

Worked Example

Example: A power station generates 200 MW of power at 25 kV. This is stepped up to 400 kV for transmission. Calculate the transmission current and compare to the original current.

```
Original: P = V × I
200 × 10⁶ = 25 × 10³ × I₁
I₁ = 200 × 10⁶ / (25 × 10³) = 8000 A (huge!)

After step-up:
200 × 10⁶ = 400 × 10³ × I₂
I₂ = 200 × 10⁶ / (400 × 10³) = 500 A (much smaller)

Ratio: 8000 A vs 500 A = 16× reduction
Power loss reduction: (8000)² vs (500)² = 64 × 10⁶ vs 0.25 × 10⁶ = 256× reduction!
```

This demonstrates why high-voltage transmission is essential: even a modest voltage step-up (from 25 kV to 400 kV, a 16× increase) reduces transmission current by 16×, dramatically reducing $I^2 R$ losses.

Exam Tip: Remember the National Grid sequence: low V at station → step-up → high V for transmission (low I, low loss) → step-down → low V for homes. The formula $P = I^2 R$ shows why high voltage is crucial: halving current quarters power loss.

Summary Table: Permanent vs Induced Magnets

Property	Permanent Magnet	Induced Magnet
Source of magnetism	Aligned atomic electrons (intrinsic)	Induced by nearby permanent magnet
Always magnetic?	Yes (unless heated above Curie temperature)	Only when near a permanent magnet
Examples	Steel, iron alloys, cobalt, magnetite	Iron, nickel, cobalt (unmagnetized)
Force with permanent magnet	Attract or repel depending on poles	Always attract (poles align automatically)
Strength control	Cannot be controlled (fixed)	Controlled by distance from permanent magnet

Summary Table: A.C. Generator vs D.C. Dynamo

Feature	A.C. Generator	D.C. Dynamo
Electrical contact type	Slip rings (two complete rings)	Split-ring commutator (two half-rings)
Output voltage type	Alternating current (sine wave)	Direct current (bumpy, rectified)

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Frequency	Same as rotation frequency (50 Hz in UK)	Pulsating d.c. at twice the frequency
Voltage waveform	Smooth sine wave (+ and − alternately)	Full-wave rectified (always positive, varying magnitude)
Common use	Power stations, mains electricity (UK grid)	Vehicles, renewable energy systems

Summary Table: Motor Effect vs Generator Effect

Aspect	Motor Effect	Generator Effect
Definition	Force on current-carrying conductor in magnetic field	Voltage induced when conductor moves in/through magnetic field
Energy conversion	Electrical → Mechanical	Mechanical → Electrical
Device example	Electric motor, loudspeaker, fan	Generator (a.c. or d.c.), microphone
Force/Voltage direction	Given by Fleming's left-hand rule	Given by Lenz's law (opposes change)
Strength depends on	B, I, L ($F = B \times I \times L$)	Rate of flux change, number of turns, coil area

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